

How to Ruin a Perfectly Good Game with Eye Tracking

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What is Reflexio?

- Puzzle game
- Made during undergrad game design class
- Reflection mechanics

Goal

- Convert game to javaScript to run in browser
- Add eye tracking to select reflection lines

- Ported C# → JavaScript
- WebGazer.js
- Webcam-based tracking
- Fully browser-based

Attempt 1: Look to Select

- Look at umbrella indicator
- Toggle reflection line
- Most straightforward approach

The First Problem

- $\sim 100\text{px}$ error
- Small targets

Possible solution: Kalman Filter

- Essentially an HMM
- Uses the last predicted state and the current observation
- Can smooth out noise

Not good enough

- Error $\sim 50\text{px}$
- Still unusable

Attempt 3: Saccades

- Use eye movement direction
- Left, right, up, down, and diagonals
- Discrete input
- Less precision required

New Problem

- Hard to repeat same saccade
- Inconsistent motion
- Opposite saccade happens
- Undoes previous action

Attempt 4: Auto Cycling

- Detect direction once
- Cycle every 2s
- Reduce repetition

User Experience

- Slow to get the reflection line you want
- Unreliable input that can toggle the wrong line
- Mentally exhausting

Comparison: Keyboard or Controller

- Instant input
- Reliable
- Allows focusing on the game, not the controls

Key Lessons

- Consistency > novelty
- Webcam eye tracking and precision input are mutually exclusive
- Bad User experience kills the gameplay

Conclusion

- Eye tracking ruined UX
- Cool idea, bad fit